Lab Exercise 20

MYSQL STRING FUNCTIONS

AIM

To implement the MySQL string functions **REPLACE, REPEAT, REVERSE, RIGHT, LEFT, RPAD, LPAD, ASCII, BIN, and OCT** using the `Vegetable` table.

PROCEDURE

1. Select the `VegetableDB` database.
2. Use `VegName` from the `Vegetable` table for string operations.
3. Apply different MySQL string functions.
4. Display the results using `SELECT` statements.

1. REPLACE()

Replaces a particular string with another string.

```
SELECT REPLACE(VegName, 'Tomato', 'Potato') AS Result
FROM Vegetable;
```

```
mysql> SELECT REPLACE(VegName, 'Tomato', 'Potato') AS Result
    -> FROM Vegetable;
+----------+
| Result   |
+----------+
| Potato   |
| Potato   |
| Carrot   |
| Onion    |
| Beetroot |
| Cabbage  |
| Spinach  |
| Beetroot |
+----------+
8 rows in set (0.00 sec)
```

2. REPEAT()

Repeats a string a specified number of times.

```
SELECT REPEAT(VegName, 2) AS Result
FROM Vegetable;
```

```
mysql> SELECT REPEAT(VegName, 2) AS Result
    -> FROM Vegetable;
+------------------+
| Result           |
+------------------+
| TomatoTomato     |
| PotatoPotato     |
| CarrotCarrot     |
| OnionOnion       |
| BeetrootBeetroot |
| CabbageCabbage   |
| SpinachSpinach   |
| BeetrootBeetroot |
+------------------+
8 rows in set (0.01 sec)
```

3. REVERSE()

Reverses the characters of a string.

```
SELECT VegName, REVERSE(VegName) AS Reversed
FROM Vegetable;
```

```
mysql> SELECT VegName, REVERSE(VegName) AS Reversed
    -> FROM Vegetable;
+----------+----------+
| VegName  | Reversed |
+----------+----------+
| Tomato   | otamoT   |
| Potato   | otatoP   |
| Carrot   | torraC   |
| Onion    | noinO    |
| Beetroot | toorteeB |
| Cabbage  | egabbaC  |
| Spinach  | hcanipS  |
| Beetroot | toorteeB |
+----------+----------+
8 rows in set (0.03 sec)
```

4. RIGHT()

Returns characters from the right side.

```
SELECT VegName, RIGHT(VegName, 3) AS Result
FROM Vegetable;
```

```
mysql> SELECT VegName, RIGHT(VegName, 3) AS Result
    -> FROM Vegetable;
+----------+--------+
| VegName  | Result |
+----------+--------+
| Tomato   | ato    |
| Potato   | ato    |
| Carrot   | rot    |
| Onion    | ion    |
| Beetroot | oot    |
| Cabbage  | age    |
| Spinach  | ach    |
| Beetroot | oot    |
+----------+--------+
8 rows in set (0.00 sec)
```

5. LEFT()

Returns characters from the left side.

```
SELECT VegName, LEFT(VegName, 3) AS Result
FROM Vegetable;
```

```
mysql> SELECT VegName, LEFT(VegName, 3) AS Result
    -> FROM Vegetable;
+----------+--------+
| VegName  | Result |
+----------+--------+
| Tomato   | Tom    |
| Potato   | Pot    |
| Carrot   | Car    |
| Onion    | Oni    |
| Beetroot | Bee    |
| Cabbage  | Cab    |
| Spinach  | Spi    |
| Beetroot | Bee    |
+----------+--------+
8 rows in set (0.00 sec)
```

6. RPAD()

Adds characters to the **right** side of a string.

```
SELECT VegName, RPAD(VegName, 10, '*') AS Result
FROM Vegetable;
```

```
mysql> SELECT VegName, RPAD(VegName, 10, '*') AS Result
    -> FROM Vegetable;
+----------+------------+
| VegName  | Result     |
+----------+------------+
| Tomato   | Tomato**** |
| Potato   | Potato**** |
| Carrot   | Carrot**** |
| Onion    | Onion***** |
| Beetroot | Beetroot** |
| Cabbage  | Cabbage*** |
| Spinach  | Spinach*** |
| Beetroot | Beetroot** |
+----------+------------+
8 rows in set (0.04 sec)
```

7. LPAD()

Adds characters to the **left** side of a string.

```
SELECT VegName, LPAD(VegName, 10, '*') AS Result
FROM Vegetable;
```

```
mysql> SELECT VegName, LPAD(VegName, 10, '*') AS Result
    -> FROM Vegetable;
+----------+------------+
| VegName  | Result     |
+----------+------------+
| Tomato   | ****Tomato |
| Potato   | ****Potato |
| Carrot   | ****Carrot |
| Onion    | *****Onion |
| Beetroot | **Beetroot |
| Cabbage  | ***Cabbage |
| Spinach  | ***Spinach |
| Beetroot | **Beetroot |
+----------+------------+
8 rows in set (0.03 sec)
```

8. ASCII()

Returns the ASCII value of the first character.

```
SELECT VegName, ASCII(VegName) AS ASCII_Value
FROM Vegetable;
```

```
mysql> SELECT VegName, ASCII(VegName) AS ASCII_Value
    -> FROM Vegetable;
+----------+-------------+
| VegName  | ASCII_Value |
+----------+-------------+
| Tomato   |          84 |
| Potato   |          80 |
| Carrot   |          67 |
| Onion    |          79 |
| Beetroot |          66 |
| Cabbage  |          67 |
| Spinach  |          83 |
| Beetroot |          66 |
+----------+-------------+
8 rows in set (0.00 sec)
```